

Stage 2 Supported Case Report Figure

Selected case: Galactose Metabolism (seed 2)

Selection basis: complete narrative | primary pathway semantic hit | 10/10 claims SUPPORTED

task_id: sub6_easy_compound_only_enrich_mammalian_RAMP_P_000000398_seed2

PRIMARY PATHWAY Galactose Metabolism KEGG:hsa00052	GOLD LABEL Galactose Metabolism semantic hit	CLAIMS 10 / 10 SUPPORTED	TOOLS CALLED 6 analysis tools
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Input metabolites and pathway members

Differential metabolite list (n=10)

Green: confirmed member. Gold: driver metabolite.

★ Galactitol

Sucrose

Glycerol

Guanabenz

Choline

myo-Inositol

Galactinol

Butanone

2-Ketobutyric acid

O-Phosphoethanolamine

Supported membership claims

✓ galactitol

✓ myo-inositol

✓ sucrose

✓ galactinol

✓ glycerol

Driver claim

★ galactitol drives KEGG:hsa00052

Enrichment evidence across tools

Primary signal reproduced by independent methods

Bar length encodes -log10(p or FDR), capped at 12.

RaMP enrichment

KEGG:hsa00052

FDR 1.64e-12

MetaboAnalyst PSEA

KEGG:hsa00052

rank 0 | p 5.1556e-08

Mummichog

galactose metabolism

rank 5 | p 0.00621796

RaMP enrichment

KEGG:hsa00564

rank 3 | FDR 3.86118e-08

Pathway predictions

Primary: Galactose Metabolism (hsa00052)

Alternative: Glycerophospholipid metabolism (hsa00564)

Claim-grounded interpretation

Evidence graph from structured claims

Edges represent member-of, driver-of, and secondary-pathway context.

Galactitol

myo-inositol

Sucrose

Galactose Metabolism

Galactinol

Glycerol

Glycerophospholipid metabolism

Final interpretation: Galactose Metabolism is the robust primary pathway because RaMP enrichment, MetaboAnalyst PSEA, and Mummichog support the same pathway; five input metabolites are confirmed members; and galactitol is identified as the driver metabolite.

Source: reports/reports_v3/stage2_detailed/2026-06-30_good_case_reports.md

Generated deterministically from structured claims